

SOURABH PORWAL

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Safety engineer with seven years in safety-critical autonomous driving, including the operational side of keeping a system safe in the field. At Vay, helped build the in-house safety function and its framework from scratch, where no teledriving regulation existed. Defined the operational design domain so the system stays safe within it, set operational speed limits by road geometry, analysed operational safety metrics such as how often the vehicles trigger a minimum-risk manoeuvre, and supported teledriver training and the guidelines they work to. Comfortable turning messy operational data and risk into clear limits, guidelines, and documentation, and building the safety processes a new market needs from the ground up.

CORE AREAS

Operational Safety & Readiness: ODD definition against system limits; operational safety metrics (MRM frequency); operational speed limits by road geometry; teledriver reaction-time analysis; operator training & guidelines support; field-incident tracking

Safety Frameworks & Process: Building safety functions and processes from scratch; safety case; safety assurance and readiness; external assessment (TÜV SÜD)

Risk Management: HARA, FMEA, FTA, FMEDA; risk assessment and mitigation; tolerable-risk thresholds

Partner & Vendor Safety: OEM and partner interface; supplier safety oversight; B2B integration

Documentation & Standards: Safety concepts (FSC / TSC), requirements and traceability (JAMA, DOORS); ISO 26262, ISO 21448

EXPERIENCE

Senior Functional Safety Engineer

09/2023 – Present

Vay Technologies GmbH · Berlin, Germany

- Defined the operational design domain (ODD) against the system's technical limits, setting the conditions under which the system stays safe in the field.
- Set operational speed limits for specific road geometries, and analysed teledriver reaction times, feeding both into operational guidelines.
- Analysed operational safety metrics, including how often vehicles trigger a minimum-risk manoeuvre, to judge whether field operations stayed within safe bounds.
- Supported teledriver training and helped define the guidelines teledrivers work to.
- Derived tolerable failure-rate thresholds for each fallback-manoevre type from probabilistic risk analysis, with an operational stop-fleet trigger if the thresholds are exceeded.
- Act as safety lead for integrating the remote-driving stack into a B2B partner's product, working directly with the partner's safety and engineering teams on integration and safety deliveries.

Functional Safety Engineer II

06/2022 – 08/2023

Vay Technologies GmbH · Berlin, Germany

- Designed the system's fail-safe fallback behaviour (Minimum Risk Manoeuvre) end to end, from concept and systems through functional safety.
- Led the functional-safety side of the independent TÜV SÜD assessment of the safety concept and risk analyses, contributing to Vay's first-in-Europe approval to operate on public roads without a safety driver.
- Performed safety analyses (FMEA, FTA) and supported verification and validation at system and vehicle level.

07/2021 – 06/2022

Functional Safety Engineer I

Vay Technologies GmbH · Berlin, Germany

- Helped establish Vay's in-house functional safety function and processes from scratch, deriving the safety framework from first principles where no tele-driving regulation existed.
- Implemented safety requirements management and traceability (JAMA), and built the safety documentation and review process for certification readiness.

Functional Safety Engineer / Manager

01/2020 – 06/2021

BERNS Engineers GmbH · Gilching, Germany

- As functional safety manager on certain projects, owned the safety plan, oversaw supplier safety activities, and provided functional-safety releases for the OEM's internal development milestones.
- Prepared safety concepts and analyses (FSC, TSC, HARA, FTA) for OEM customers and worked directly with their architects on ASIL D ADAS features.

Functional Safety Engineer

06/2019 – 12/2019

AKKA Technologies · Ingolstadt, Germany

- Supported ISO 26262 development for an ASIL B system and managed vehicle configuration data.

Junior Executive

08/2013 – 09/2015

Fleetguard Filters Pvt. Ltd. · Pune, India

- Managed production part approval process (PPAP) activities across manufacturing suppliers, including process FMEA and process flow documentation; reduced component cost by ~25% through supplier negotiations.

EDUCATION

M.Sc. Mechanical Engineering

10/2015 – 04/2019

Rhine-Waal University of Applied Sciences · Kleve, Germany

B.E. Mechanical Engineering

08/2009 – 04/2013

Visvesvaraya Technological University · Belgaum, India

CERTIFICATION

CertX – ISO 26262:2018 Functional Safety Red Belt (03/2021)

PERSONAL DETAILS

Mobility: Open to relocation

Languages: English (fluent), German (B1), Hindi (native)